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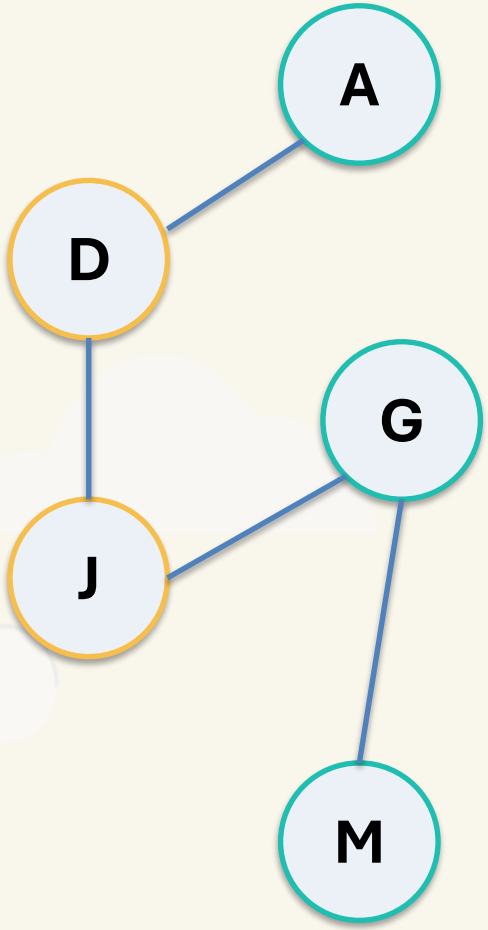
Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**CSA5112 - CRYPTOGRAPHY AND NETWORK SECURITY**

**CO1 AT3 - Technical Presentation**

# From Caesar to Vigenère : Evolution of Substitution Ciphers




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# Substitution ciphers

In a substitution cipher, each plaintext letter is replaced by another letter.

**PLAINTEXT**

**The original readable message**

**Example: HELLO**

**ENCRYPTION**

**Apply a key-based letter mapping**

**A → D, B → E, …**

**CIPHERTEXT**

**The transformed message**

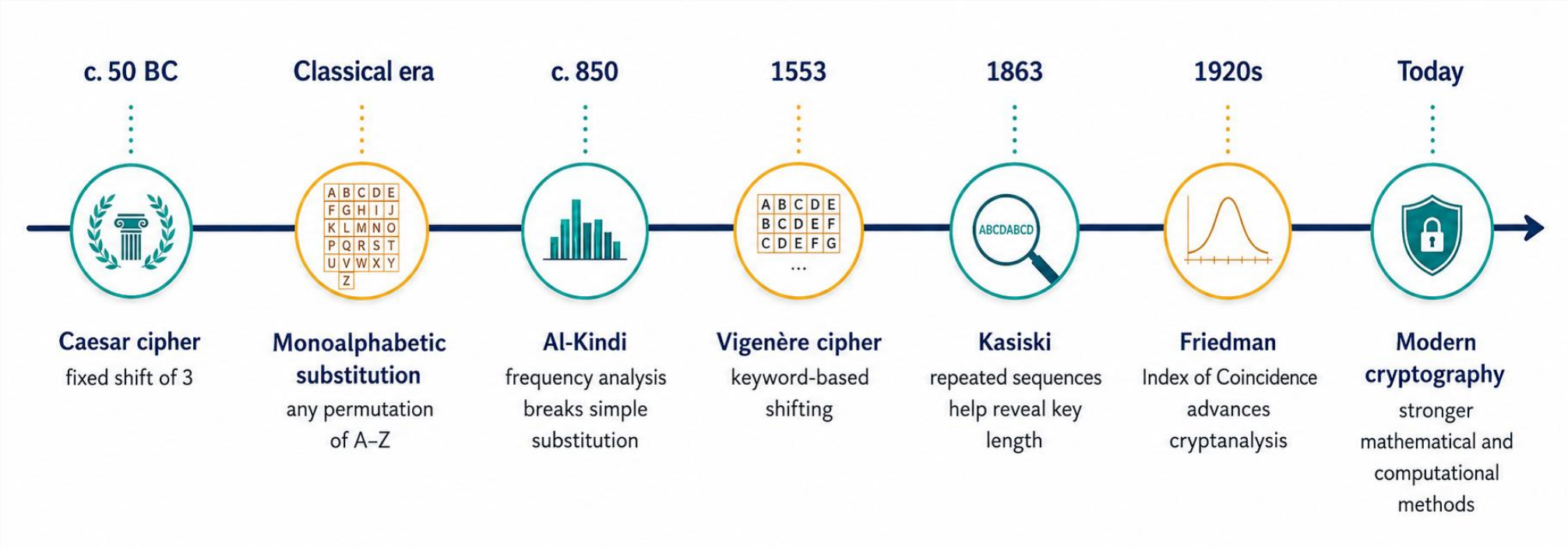
**Example: KHOOR**

## Core goals

- **Confidentiality:** hide the message from unauthorized readers.
- **Reversibility:** the intended receiver uses the key to recover the original text.
- **Important distinction:** substitution changes letters; transposition changes their positions.



# How the methods developed




**Fixed shift → fixed substitution alphabet → multiple alphabet shifts → modern computational cryptography**

# Caesar cipher



Every plaintext letter is shifted by the same number of positions.

**ENCRYPTION RULE**

$$C = (P + k) \bmod 26$$

P = plaintext index, C = ciphertext index, k = shift

**DECRYPTION RULE**

$$P = (C - k) \bmod 26$$

Use A = 0, B = 1, …, Z = 25

**Worked calculation • key k = 3**

| **Plaintext** | H | E | L | L | O |
|---|---|---|---|---|---|
| **Index P** | 7 | 4 | 11 | 11 | 14 |
| **(P + 3) mod 26** | 10 | 7 | 14 | 14 | 17 |
| **Ciphertext** | **K** | **H** | **O** | **O** | **R** |

**Result: HELLO → KHOOR | Decryption: KHOOR → HELLO**

# Vigenère cipher

The keyword is repeated to decide the shift used for each plaintext letter.

**ENCRYPTION**

$$C_i = (P_i + K_i) \bmod 26$$

**Decryption:** $P_i = (C_i - K_i) \bmod 26$

**KEY ALIGNMENT**

**Plaintext:** ATTACKATDAWN

**Key:** LEMONLEMONLE

## Worked calculation • A = 0, …, Z = 25

| Letter | A | T | T | A | C | K |
|---|---|---|---|---|---|---|
| P index | 0 | 19 | 19 | 0 | 2 | 10 |
| Key letter | L | E | M | O | N | L |
| K index | 11 | 4 | 12 | 14 | 13 | 11 |
| (P + K) mod 26 | 11 | 23 | 5 | 14 | 15 | 21 |
| Cipher | **L** | **X** | **F** | **O** | **P** | **V** |

**Result: ATTACKATDAWN → LXFOPVEFRNHR**

# Procedure and test cases

**1. INPUT**
Plaintext + key

**2. NORMALIZE**
Letters A–Z
keep key valid

**3. ENCRYPT**
Apply formula
letter by letter

**4. DECRYPT**
Reverse formula
using same key

**5. VERIFY**
Recovered text
matches input

- *Test Case 1*

  ***Caesar:***

  **plaintext –** HELLO ; **key** – 3 ; **expected ciphertext -** KHOOR.



- *Test Case 2*

  ***Vigenère:***

  **plaintext** – ATTACKATDAWN ; **key** – LEMON ; **expected ciphertext -** LXFOPVEFRNHR.

- Decrypt each result and compare it with the original plaintext.
- For fair comparison use the same alphabet mapping (A = 0 to Z = 25) and document spaces/punctuation handling.

# Comparison of Caesar and Vigenère

| Feature | Caesar | Vigenère |
|---|---|---|
| **Key** | One shift number | Keyword repeated across text |
| **Letter mapping** | Same shift each time | Shift varies by key position |
| **Key space** | 26 possible shifts | Much larger, depending on key length |
| **Pattern leakage** | Very high | Reduced when key is long and well chosen |
| **Typical attack** | Try all 26 shifts | Frequency analysis / repeated-key methods |
| **Modern security** | Insecure | Insecure if used as a classical cipher |

**A more complicated classical cipher is not automatically a secure cipher.**

# Results and observations

**RESULT 1**

**HELLO → KHOOR**

A fixed shift is easy to reverse by testing the 26 possible keys.

**RESULT 2**

**ATTACKATDAWN → LXFOPVEFRNHR**

The key changes shifts across positions, masking simple letter frequencies more effectively.

**VALIDATION**

**Decrypt → original**

Correct recovery confirms that the formulas and key alignment were applied consistently.

## Threats and limitations

- Short messages provide limited evidence; test several messages and key choices for stronger evaluation.
- Vigenère can leak patterns when a keyword repeats; key length and language statistics can be exploited.
- These ciphers are educational examples—not suitable for protecting real personal, financial, or business data.

# Conclusion

- Caesar demonstrates a simple, fixed shift.
- Vigenère uses a keyword to apply changing shifts across the message.
- The worked examples were checked by decrypting the ciphertext.
- Both methods are vulnerable to systematic analysis and should not protect real data.

**Complexity is not the same as security.**

Modern protection requires well-reviewed cryptographic algorithms and sound key management.